

General Science

I. PHYSICS

Measurement/Unit

1. In the following which is fundamental physical quantity?

- (a) Force
- (b) Velocity
- (c) Electric current
- (d) Work
- (e) None of the above/More than one of the above

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Ans. (c)

In physics, there are seven fundamental physical quantities (which are measured in base or fundamental units) : Length, Mass, Time, Electric current, Temperature, Luminous intensity and Amount of a substance.

2. The unit of electric power is :

- (a) Ampere
- (b) Volt
- (c) Coulomb
- (d) Watt
- (e) None of the above / More than one of the above

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64th B.P.S.C. (Pre) 2018

Ans. (d)

Electric power is the rate per unit time, at which electrical energy is transferred by an electric circuit. The SI unit of electric power is Watt. Ampere is the unit of electric current, Volt is the unit of electrical potential and Coulomb is the unit of electric charge in the SI system.

3. At the time of a short circuit, the current in the circuit

- (a) varies continuously
- (b) reduces substantially
- (c) does not change
- (d) increases heavily

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Ans. (d)

The potential difference/voltage in the domestic circuit is provided by live and neutral wire. When due to any reason, live wire and neutral wire get connected to each other i.e. short circuit, a huge amount of current starts flowing in the circuit because the path of current has shortened and a low resistance path for current has been made. That's why at the time of short circuit the current in the circuit increases heavily. To stop the overloading of current in the circuit, electric fuse is used.

4. Frequency is measured in :

- (a) hertz
- (b) metre/second
- (c) radian
- (d) watt
- (e) None of the above / More than one of the above

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Ans. (a)

The Hertz (symbol : Hz) is the derived unit of frequency in the International System of Units (SI system) and is defined as one cycle per second.

5. What is measured in hertz?

- (a) Frequency
- (b) Energy
- (c) Heat
- (d) Quality
- (e) None of the above/More than one of the above

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Ans. (a)

See the explanation of the above question.

6. Bolometer is used to measure

- (a) frequency
- (b) heat radiations
- (c) velocity
- (d) wavelength
- (e) None of the above / More than one of the above

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Ans. (b)

A bolometer is used to detect Infrared rays. A bolometer is a device used for detecting and measuring the heat and radiation of microwave energy.

7. The hearing frequency range of cat is

- (a) 20 Hz to 20 kHz
- (b) 20 Hz to 50 kHz
- (c) 45 Hz to 64 kHz
- (d) 10 Hz to 100 kHz
- (e) None of the above / More than one of the above

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Ans. (*)

The hearing range of the cat for sounds of 70 dB SPL extends from 48 Hz to 85 kHz, giving it one of the broadest hearing ranges among mammals. Analysis suggests that cats evolved extended high-frequency hearing without sacrificing of low-frequency hearing.

8. 'Ohm-meter' is unit of :

- (a) Resistance (b) Conductance
(c) Resistivity (d) Charge
(e) None of the above/More than one of the above

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Ans. (c)

Electrical resistivity is an intrinsic property that quantifies how strongly a given material opposes the flow of electric current. A low resistivity indicates a material that readily allows the movement of electric charge. The SI unit of electrical resistivity is ohm-meter (Ωm). It is commonly represented by the-

Greek letter ρ (rho) defined as $\left[\rho = \frac{RA}{l}\right]$

Here,

R = electrical resistance of the material

l = Length, A = Cross section area, ρ = resistivity.

9. 'Light-year' is a unit of :

- (a) Time (b) Distance
(c) Speed (d) Force
(e) None of the above/More than one of the above

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Ans. (b)

Light-year is a unit of astronomical distance equivalent to the distance that light travels in one year, which is 9.46×10^{15} metres.

10. Angstrom is a unit of

- (a) Wavelength (b) Energy
(c) Frequency (d) Velocity
(e) None of the above / More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Angstrom ($\text{\AA}$) is the unit of length used mainly for measuring the wavelengths of electromagnetic waves which is equal to 10^{-10} meter or 0.1 nanometer. It is named after the 19th century Swedish physicist Anders Jonas Angstrom.

11. 1 zepto is equal to

- (a) 10^{-25} (b) 10^{25}
(c) 10^{-21} (d) 10^{-22}
(e) None of the above / More than one of the above

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Ans. (c)

In metric system, 1 zeptometer = 10^{-21} meter.

12. 1 femtometer is equal to how many meters?

- (a) 10^{-12} m (b) 10^{-14} m
(c) 10^{-15} m (d) 10^{-24} m
(e) None of the above / More than one of the above

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Ans. (c)

1 femtometer is equal to 10^{-15} meters. The symbol of femtometer is Fm.

13. One micron is equal to –

- (a) 1/10 mm (b) 1/100 mm
(c) 1/1000 mm (d) 1/10,000 mm

39th B.P.S.C. (Pre) 1994

Ans. (c)

$$\begin{aligned} 1 \text{ micron} &= 10^{-6} \text{ m} \\ &= 10^{-6} \times 10^3 \text{ mm} \\ &= \frac{1}{10^3} \text{ mm} \\ &= \frac{1}{1000} \text{ mm} \end{aligned}$$

14. A micron is related to centimetre as

- (a) 1 micron = 10^{-4} cm (b) 1 micron = 10^{-8} cm
(c) 1 micron = 10^{-6} cm (d) 1 micron = 10^{-5} cm

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Ans. (a)

$$\begin{aligned} 1 \text{ micron } (\mu\text{m}) &= 10^{-6} \text{ m} \\ \because 1 \text{ m} &= 100 \text{ cm} \\ \therefore 1 \mu\text{m} &= 10^{-6} \times 10^2 \text{ cm} \\ 1 \mu\text{m} &= 10^{-4} \text{ cm} \end{aligned}$$

15. A distance of 1 km means –

- (a) 100 m (b) 1000 cm
(c) 1000 m (d) 100 cm

45th B.P.S.C. (Pre) 2001

Ans. (c)

The SI unit for distance is meter. 1 km is equal to 1000m. 1 m is equal to 100 cm.

16. One picogram is equal to –

- (a) 10^{-6} gram (b) 10^{-9} gram
(c) 10^{-12} gram (d) 10^{-15} gram

42nd B.P.S.C. (Pre) 1997

Ans. (c)

The smaller units of measuring mass are Milligram, Microgram, Picogram and Femtogram etc.

$$\begin{aligned} 1 \text{ Picogram} &= 10^{-12} \text{ gram} & 1 \text{ Milligram} &= 10^{-3} \text{ gram} \\ 1 \text{ Microgram} &= 10^{-6} \text{ gram} & 1 \text{ Nanogram} &= 10^{-9} \text{ gram} \\ 1 \text{ Femtogram} &= 10^{-15} \text{ gram} \end{aligned}$$